Marsupials

PHYLUM	Chordata
CLASS	Mammalia
INFRACLASS	Marsupialia
ORDERS	7
FAMILIES	19
SPECIES	363

CLASSIFICATION NOTE

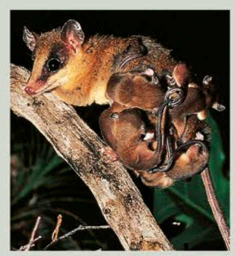
The taxonomic rank of Marsupialia has changed. It is now an infraclass containing 7 orders: Didelphinomorphia (the American opossums), Dasyuromorphia (the Australian carnivorous marsupials), Peramelemorphia (bandicoots and bilbies), Notoryctemorphia (marsupial "moles"), Diprotodontia (kangaroos, koala, and relatives), Paucituberculata ("shrew" opossums), and Microbiotheria (the monito del monte). Because of their shared reproductive biology, the orders are not being treated separately but as infraclass Marsupialia.

Like other mammals (apart from the monotremes), marsupials bear live offspring. They are, however, distinct from all other live-bearers (together described as placental mammals), in that they give birth at a very early stage of the embryo's development and nourish the newborn on milk rather than by a placenta. Marsupials are, therefore, classified as infraclass Marsupialia (or Metatheria), while the placental mammals are placed in the infraclass Eutheria. Marsupials are amazingly diverse, including animals such as kangaroos, possums, and bandicoots. The Australasian marsupials, through a lack of competing species, have diversified and become specialized insectivores, carnivores, and herbivores. In South America, the marsupials are small and mostly arboreal; only one species, the Virginia opossum, has spread to North America.

Early life

Marsupial offspring are born in an almost embryonic state after a very short gestation (only 12 days in some bandicoot species). The newborn makes its way to one of the mother's nipples, where it remains attached for several weeks. Larger species have single births, but the small quolls and dunnarts have litters of up to 8. In many kangaroo species, the female mates again while pregnant, but the new embryo remains dormant until the previous young leaves the pouch.

AMAZING JOURNEY
This tiny tammar wallaby baby is hauling itself over its mother's fur to reach a nipple, to which it will firmly attach itself. As in most marsupials, the nipples are protected by a pouch. The joey will not relinquish the nipple until, months later, it begins to explore the outside world.



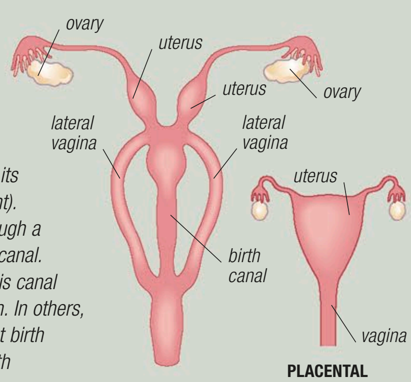
OVERCROWDING
The mother's pouch may be too small to carry numerous offspring as they grow and so she has no option but to carry her babies externally.

Anatomy

Externally, marsupials are highly varied, although many have long back legs and feet (for example, kangaroos) and elongated snouts; almost all have large eyes and ears. Female marsupials have a unique "doubled" reproductive tract (see below), and in some males the penis is forked. The testes are held in a pendulous scrotum with a long, thin stalk, which swings in front of the penis. Apart from their specialized reproductive system, marsupials also differ from placental mammals in that their brain is relatively smaller and lacks a corpus callosum (the nerve tract connecting the two cerebral hemispheres). The group that contains the kangaroos, possums, wombats, and the koala, and the group containing the bandicoots, have an arrangement called syndactyly: the second and third toes of the back foot are combined to form a single digit with 2 claws.

SPECIAL POUCHES
Most marsupials have a pouch to carry their young. This western gray kangaroo has a forward-facing pouch, from which her joey's head is protruding. Some pouches face backward.

FEMALE REPRODUCTIVE SYSTEM
Unlike placental mammals, which have a single uterus and vagina (far right), female marsupials have a double system with 2 uteri, each with its own lateral vagina (right). The young is born through a separate, central birth canal. In some marsupials, this canal forms before each birth. In others, it remains after the first birth (but sometimes fills with connective tissue).



Movement

Although most marsupials run or scurry, there are several variations. For example, wombats waddle, while koalas and possums climb. Kangaroos and wallabies hop on their long back legs, using the extended middle toe as an extra limb segment. Although hopping at low speeds uses more energy than running on all fours, above approximately 6 ft (1.8 m) per second, the larger species begin to conserve energy. This is because energy is stored in the tendons of the foot, and the heavy tail swings up and down like a pendulum, providing momentum.

HOPPING
The faster a large kangaroo (such as this red kangaroo) moves, the more energy efficient it becomes.

GLIDING
Some possums, known as gliders, have a membrane between their front and back legs, which they use as a parachute.